

BRG-Bokeh: Boundary-Refined 3DGS Segmentation for Multi-View Consistent Bokeh

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Abstract—Generating consistent bokeh effects for foreground objects in 3D scenes across multiple viewpoints poses a significant challenge for single-image bokeh methods, due to inconsistent focus target localization and prominent boundary artifacts. To address these challenges, we propose a novel framework that integrates depth-regularized scene reconstruction and boundary-refined 3D segmentation to learn a 3D Gaussian Splatting (3DGS) representation. Specifically, we leverage Depth Anything v3 to generate monocular depth maps and SAM3 to produce text-prompt-driven masks, which respectively serve as ground truth for depth estimation and segmentation. This allows the learned 3DGS to simultaneously generate RGB images, depth maps, and spatially consistent masks of the target object across multiple views. Furthermore, we develop a multi-view consistent bokeh rendering method that reassigns depth values to pixels within the in-focus object and applies dual-band Gaussian smoothing to mitigate edge artifacts. Comprehensive experiments on segmentation and bokeh datasets demonstrate that our approach outperforms 3D segmentation methods and single-image bokeh methods. Notably, our method realizes flexible, natural, and multi-view consistent bokeh rendering, providing a promising solution for scenarios requiring highly precise and coherent defocus effects across multiple viewpoints. The source code is available at <https://github.com/jiutaojushi/BRG-Bokeh>.

Index Terms—Multi-view consistent bokeh, 3D Gaussian Splatting, 3D Segmentation, Foundation Models, Depth of Field

I. INTRODUCTION

Bokeh achieves Depth of Field (DoF) effect of a Digital Single-Lens Reflex (DSLR) camera by blurring the regions beyond the focal plane, which can improve the aesthetic effect of the image and highlight the main subjects by changing different lens’s parameters and in-focus object.

Early approaches to synthesizing DoF focused on simulating physical optics. Methods such as that proposed by Bertalmio et al. [1] leveraged anisotropic diffusion partial differential equation to approximate the optical phenomenon camera for producing real-time, accurate DoF effects. SteReFo [2] leverages disparity maps derived from stereo input to generate multiple images focused at different depths and a final refocused image. For single-camera mobile platforms, researchers developed methods that synthesize DoF by estimating depth from a single image [3], [4], while others integrated semantic understanding using salient object detection to guide the application of shallow DoF for better artistic results [5]. Dr.Bokeh

[6] presented filtering-based bokeh rendering equation and a physically-based occlusion-aware bokeh renderer to address color bleeding and partial occlusion at object boundaries.

The advent of deep learning has revolutionized bokeh generation, shifting the paradigm from purely physically based simulation to data-driven synthesis. Ignatov et al. [7] introduced the EBB! dataset containing 10,000 images and proposed an inverted pyramidal shape network for predicting bokeh images end-to-end. This was followed by more specialized neural architectures, such as BGGANs [8], which employed generative adversarial networks to produce realistic, lens-specific bokeh shapes [8], and multi-scale hierarchical networks optimized for fast rendering from a single image [9]. Addressing artifacts related to partial occlusion, Peng et al. [10] proposed multi-plane image-based bokeh to implement high-resolution scene representation and to render various bokeh effects. Peng et al. [11] presented a ray-tracing-based bokeh simulator to create training data for learning a hybrid framework by combining a classic renderer and a neural renderer.

The integration of DoF and bokeh effects into Neural Radiance Fields (NeRFs) [12] and 3D Gaussian Splatting (3DGS) [13] has further enabled novel view bokeh generation. Pioneer works like DoF-NeRF [14] and NeRFocus [15] modify volume rendering to simulate camera optics, generating user-controllable defocus effects within 3D scene representations. Similarly, in the context of 3DGS, methods like DoF-Gaussian [16] and DOF-GS [17] have been proposed to simulate aperture effects and focal plane adjustments within the differentiable splatting pipeline. Huang et al. [18] utilized 3DGS and proposed a framework for multi-view consistent bokeh by tracking salient objects and segmenting them with SAM2 [19]. However, saliency detection fails to flexibly specify user-desired in-focus objects, and multi-stage operations cannot guarantee consistent object segmentation.

In this paper, we aim to generate multi-view consistent bokeh effects for images captured from the same scene. We propose a framework integrating depth-regularized scene reconstruction and boundary-refined 3D segmentation to learn a 3DGS representation [13], which can simultaneously generate RGB images, depth maps, and masks of the target object. Specifically, we leverage Depth Anything v3 (DA3) [20] to generate monocular depth maps and SAM3 [21] to produce text-prompt-driven masks, which serve as pseudo-ground truth for depth estimation and segmentation, respectively. Conse-

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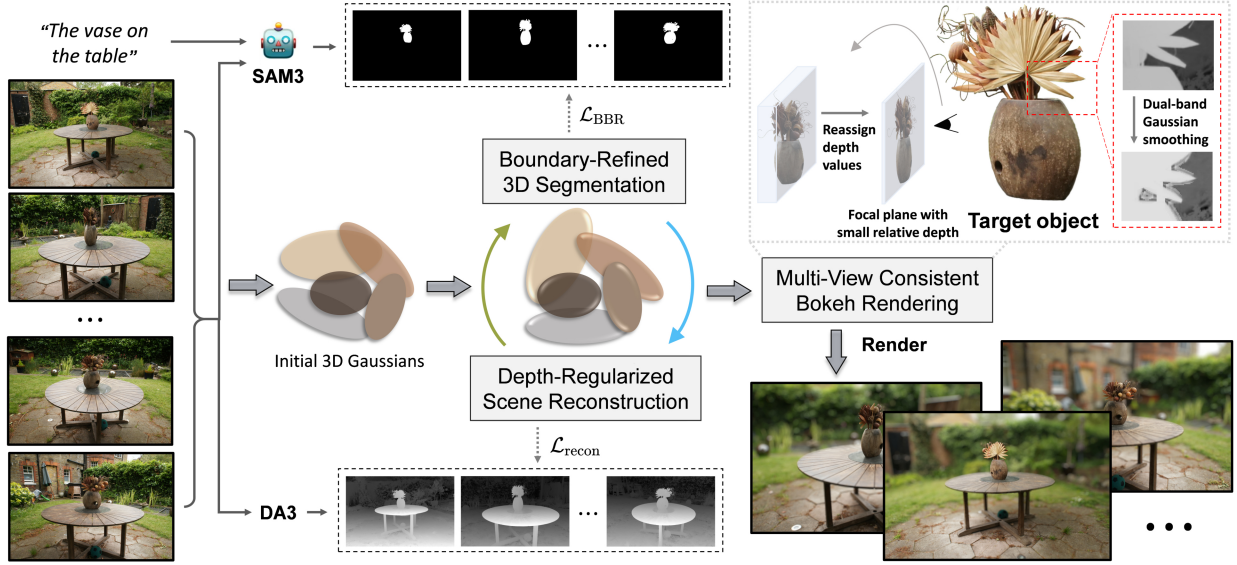


Fig. 1. The main framework of the proposed method. “DA3” denotes Depth Anything v3.

quently, the learned 3DGS can accurately mask the desired in-focus object across multiple views, thereby guaranteeing the consistency of bokeh effects. Besides, we propose a multi-view consistent bokeh rendering method that reassigns depth values of pixels within the in-focus object and applies dual-band Gaussian smoothing to mitigate edge artifacts. We have conducted comprehensive experiments on relevant segmentation and bokeh datasets. Our method supports text-based specification of the in-focus object, offering greater flexibility than saliency detection, outperforming other 3D segmentation methods in segmentation quality, and enabling flexible, natural, and multi-view consistent bokeh rendering. The main contributions are as follows:

- We propose a 3DGS framework integrating depth-regularized reconstruction and boundary-refined 3D segmentation, which simultaneously generates RGB images, depth maps, and in-focus masks while ensuring cross-view consistent target masking.
- We propose a multi-view consistent bokeh rendering method that reassigns depth values of in-focus object pixels and integrates dual-band Gaussian smoothing. It effectively reduces edge artifacts, enabling natural and coherent cross-view bokeh.
- Our method supports text-based in-focus object specification, outperforms existing 3D segmentation methods, and achieves more flexible, natural, and multi-view-consistent bokeh compared to single-image bokeh methods.

II. PRELIMINARY

A. 3D Gaussian Splatting (3DGS)

3D Gaussian Splatting [22] represents 3D scenes through a collection of learnable Gaussian primitives $\mathcal{G} = \{G_i\}$. Each primitive $G_i = \{\mu_i, \mathbf{s}_i, \mathbf{q}_i, o_i, \mathbf{c}_i\}$ is characterized by its position $\mu_i \in \mathbb{R}^3$, scale $\mathbf{s}_i \in \mathbb{R}^3$, rotation quaternion $\mathbf{q}_i \in \mathbb{R}^4$, opacity o_i , and spherical harmonic coefficients $\mathbf{c}_i$ for view-dependent color. During rendering, 3D Gaussians are projected

onto the image plane to form 2D splats. The pixel color c is computed via differentiable alpha blending:

$$c = \sum_{i \in \mathcal{N}} c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \quad (1)$$

where c_i denotes the view-dependent color. The blending weight α_i is determined by the learned opacity o_i modulated by the 2D Gaussian probability evaluated at pixel $\mathbf{x}$:

$$\alpha_i = o_i \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}')^T \Sigma'^{-1} (\mathbf{x} - \boldsymbol{\mu}') \right), \quad (2)$$

where $\boldsymbol{\mu}'$ and Σ' represent the center and covariance matrix of the Gaussian projected onto the 2D image plane, respectively. An adaptive density control mechanism dynamically manages the Gaussian population, enabling reconstruction from sparse initializations to detailed scene representations.

B. Bokeh Rendering

Bokeh rendering simulates depth-of-field effects via the circle of confusion model. Under thin-lens geometry, a scene point at depth z_p deviating from the focal plane z_f produces a blur disk with radius:

$$r = \kappa \cdot L \cdot f \cdot \left| \frac{1}{z_p} - \frac{1}{z_f} \right|, \quad (3)$$

where κ is a configurable scaling factor, L is the aperture size and f is the focal length. Given RGB image $\mathbf{I}$, depth map $\mathbf{D}$, mask $\mathbf{M}$ (for handling occlusion and boundary artifacts), and lens parameters k , the bokeh output is:

$$\mathbf{B} = \phi_{\text{Bokeh}}(\mathbf{I}, \mathbf{D}, \mathbf{M}, z_f, k). \quad (4)$$

In this paper, we adopt Dr.Bokeh [6] as bokeh generator $\phi_{\text{Bokeh}}()$, which incorporates geometric occlusion handling for physically accurate rendering.

III. METHOD

Given a set of multi-view images $\mathcal{I} = \{\mathbf{I}_1, \mathbf{I}_2, \dots, \mathbf{I}_K\}$ captured from a scene, our goal is to synthesize a bokeh image $\mathbf{B}_v$ for the v -th view direction by Eq.(4). Note that the in-focus objects in $\mathbf{B}_{v_1}$ and $\mathbf{B}_{v_2}$ should be identical for arbitrary views v_1 and v_2 . This requires that we can accurately segment the same objects in different rendered images. To achieve this, we learn a 3DGS to render a RGB image $\mathbf{I}$, a depth map $\mathbf{D}$, and a mask $\mathbf{M}$ for a given view matrix $V = (R, T)$, where R is a rotation matrix, and T is a translation matrix. The rendering process can be represented as $\mathbf{I}, \mathbf{D}, \mathbf{M} = \mathcal{S}(\mathcal{G}, V)$. Fig. 1 illustrates the overall pipeline of the proposed method.

A. Depth-Regularized Scene Reconstruction

Accurate depth estimation is essential for both semantic segmentation and bokeh rendering. Inaccurate depth induces semantic ambiguity along viewing rays: foreground and background Gaussians become entangled in their alpha-composited pixel contributions, which impairs subsequent mask optimization and defocus rendering. To guarantee geometric consistency across multiple views, we incorporate monocular depth priors to regularize the 3D structure during reconstruction. For generating pixel-wise depth, we utilize alpha blending on the z -buffer obtained from the ordered Gaussians. The depth d of a given pixel is calculated as:

$$d = \sum_{i=1}^N d_i \alpha_i T_i, \quad (5)$$

where d_i denotes the z -buffer value of the i -th Gaussian.

To enforce geometric consistency, we minimize the L_1 distance between the rendered depth $\mathbf{D}^v$ and the pseudo-ground truth depth maps $\mathbf{D}_{\text{GT}}^v$ as:

$$\mathcal{L}_{\text{depth}} = |\mathbf{D}^v - \mathbf{D}_{\text{GT}}^v|_1. \quad (6)$$

The overall reconstruction loss combines photometric and depth regularization as:

$$\mathcal{L}_{\text{recon}} = (1 - \lambda_{\text{dssim}}) \mathcal{L}_1 + \lambda_{\text{dssim}} \mathcal{L}_{\text{D-SSIM}} + \lambda_{\text{depth}} \mathcal{L}_{\text{depth}}, \quad (7)$$

where $\mathcal{L}_1$ and $\mathcal{L}_{\text{D-SSIM}}$ are standard photometric losses from vanilla 3DGS [22]. We set $\lambda_{\text{depth}} = 0.005$ and $\lambda_{\text{dssim}} = 0.2$ across all experiments. Depth regularization ensures geometric consistency across multiple views, which disentangles foreground and background Gaussians along viewing rays and enhances separability to facilitate subsequent mask optimization. Furthermore, precise depth is critical for bokeh rendering (Sec. III-D), as focal plane estimation and defocus radius calculation rely directly on the correctness of the geometric structure.

B. Boundary-Refined 3D Segmentation

We augment each Gaussian primitive G_i with a learnable scalar mask logit, which is transformed into a probability $m_i \in [0, 1]$ via a sigmoid function. The 2D segmentation mask $\mathbf{M}$ is rendered similarly to color:

$$\mathbf{M}(\mathbf{x}) = \sum_{i \in \mathcal{N}} m_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j). \quad (8)$$

To supervise the mask label we propose a boundary and background regularized loss:

$$\mathcal{L}_{\text{BBR}} = \mathcal{L}_{\text{base}} + \lambda_{\text{edge}} \mathcal{L}_{\text{edge}} + \lambda_{\text{bg}} \mathcal{L}_{\text{bg}}, \quad (9)$$

where $\mathcal{L}_{\text{base}}$, $\mathcal{L}_{\text{edge}}$ and $\mathcal{L}_{\text{bg}}$ are basic mask projection loss, boundary-weighted focal-tversky loss and background area penalty loss, respectively.

$\mathcal{L}_{\text{base}}$ is computed over all pixels $\mathcal{P}$ as:

$$\mathcal{L}_{\text{base}} = \frac{1}{|\mathcal{P}|} \sum_{u \in \mathcal{P}} \left[-\mathbf{M}_{\text{GT}}(u) \mathbf{M}(u) + (1 - \mathbf{M}_{\text{GT}}(u)) \mathbf{M}(u) \right], \quad (10)$$

where $\mathbf{M}_{\text{GT}}$ is the pseudo-ground truth mask derived from SAM3 [21].

$\mathcal{L}_{\text{edge}}$ is used to improve the boundaries of the objects, which is defined as:

$$\mathcal{L}_{\text{edge}} = \frac{1}{|B|} \sum_{u \in B} (1 - \text{TI}(\mathbf{M}(u), \mathbf{M}_{\text{GT}}(u)))^\gamma, \quad (11)$$

where $B = \text{Dilate}_r(\mathbf{M}_{\text{GT}}) - \text{Erode}_r(\mathbf{M}_{\text{GT}})$ is a morphological boundary band of $\mathbf{M}_{\text{GT}}$. $\text{TI}(\mathbf{M}, \mathbf{M}_{\text{GT}}) = \frac{\mathbf{M} \cdot \mathbf{M}_{\text{GT}}}{\mathbf{M} \cdot \mathbf{M}_{\text{GT}} + \alpha_T \mathbf{M} \cdot (1 - \mathbf{M}_{\text{GT}}) + \beta_T \mathbf{M}_{\text{GT}} \cdot (1 - \mathbf{M})}$ is tversky index, where α_T and β_T are balance parameters of false positives and false negatives, respectively. This loss steers gradients toward Gaussians whose projections overlap the boundary band, enabling targeted refinement.

$\mathcal{L}_{\text{bg}}$ is a global penalty on the total predicted foreground mass in background regions to reduce background misclassification, which is defined as:

$$\mathcal{L}_{\text{bg}} = \frac{\sum_{u \in \mathcal{P}} \mathbf{M}(u) \cdot (1 - \mathbf{M}_{\text{GT}}(u))}{\sum_{u \in \mathcal{P}} (1 - \mathbf{M}_{\text{GT}}(u)) + \epsilon}, \quad (12)$$

where $\mathcal{P}$ denotes the set of all pixels in the image, and $\epsilon = 10^{-6}$ is a smoothing constant for numerical stability. We also adopt boundary-adaptive Gaussian splitting [23] to alleviate semantically ambiguous boundary Gaussians. Gaussians with large mask gradients $|\nabla m_i|$ exceeding a threshold τ_{split} are split into two smaller Gaussians with slightly perturbed positions and reduced scales.

We empirically set $\lambda_{\text{edge}} = 0.2$, $\lambda_{\text{bg}} = 5.0$, $r = 2$ pixels, $\alpha_T = 0.3$, $\beta_T = 0.7$, and $\gamma = 2$. For the splitting threshold, we set $\tau_{\text{split}} = 0.5$ for forward-facing scenes and $\tau_{\text{split}} = 0.8$ for surrounding scenes.

C. Learning 3DGS for bokeh

Directly learning the 3DGS representation of $\{\mu_i, \mathbf{s}_i, \mathbf{q}_i, o_i, \mathbf{c}_i, m_i\}$ results in conflicting gradients. Specifically, mask supervision may pull geometry in ways that degrade color fidelity, while photometric loss may distort semantics. Thus, we propose to learn the 3DGS representation by jointly conducting depth-regularized scene reconstruction (DRSC) and boundary-refined 3D segmentation (BR3DS). We first learn $\{\mathbf{x}_i, \mathbf{s}_i, \mathbf{q}_i, o_i, \mathbf{c}_i\}$ without m_i via the DRSC loss (Eq. 7) for 30,000 iterations to generate an initial 3DGS representation. Subsequently, we perform alternating

optimization for a total of $22 \times K$ iterations, where K is the number of training views. We switch the optimization objective every $2 \times K$ iterations, alternating between updating the scene attributes and refining the masks. This alternation enables masks and appearances to evolve coherently without direct mutual interference.

D. Multi-View Consistent Bokeh Rendering

Given a set of images $\mathcal{I} = \{\mathbf{I}_1, \mathbf{I}_2, \dots, \mathbf{I}_K\}$ captured from a scene, we employ Depth Anything v3 [20] to derive a monocular depth set $\mathcal{D} = \{\mathbf{D}_1, \mathbf{D}_2, \dots, \mathbf{D}_K\}$ and utilize SAM3 [21] to generate a mask set $\mathcal{M} = \{\mathbf{M}_1, \mathbf{M}_2, \dots, \mathbf{M}_K\}$ based on a specific text prompt. SAM3 [21] features Promptable Concept Segmentation (PCS), allowing us to identify target regions using a single textual prompt, which ensures the semantic consistency of the target concept throughout $\mathcal{I}$. Utilizing $\mathcal{I}$, $\mathcal{D}$, and $\mathcal{M}$, we learn a robust 3DGS representation $\mathcal{G}$, capable of simultaneously rendering RGB images, depth maps, and masks for novel viewpoints.

Let $\mathbf{I}_v, \mathbf{D}_v, \mathbf{M}_v = \mathcal{S}(\mathcal{G}, V)$ denote the rendering function for a target view matrix V . The final bokeh image is generated via Eq. (4) given lens parameters k and a focal plane distance z_f . Since z_f governs the depth of the sharpest focus, it must be adaptively adjusted for each view based on the corresponding depth map. This ensures that the desired object remains consistently sharp across all viewing angles.

According to the circle of confusion (CoC) formula Eq.(3), the radius r becomes zero when the depth of the scene point z_p equals to the focal plane distance z_f . Thus, the regions with depth of z_f are sharp. That is to say, we can set z_f according to the depth of the target, i.e., $z_f = \max(\mathbf{M}_v \cdot \mathbf{D}_v)$. We reassign the depth value of $\mathbf{D}_v$ as:

$$\mathbf{D}'_v(u) = \begin{cases} z_f & \text{if } u \in \Omega \\ \mathbf{D}_v(u) & \text{otherwise} \end{cases}. \quad (13)$$

However, uniform depth allocation gives rise to an unnatural flatness in appearance. To retain the natural surface undulations, we propose a depth perturbation derived from the original depth distribution. Specifically, we compute the depth offset of each pixel relative to z_f , scale this offset by a factor t , and superimpose the result onto the assigned focal depth to restore the front-to-back relative spatial hierarchy:

$$\mathbf{D}''_v(u) = z_f + t \cdot (\mathbf{D}_v(u) - z_f), \quad u \in \Omega, \quad (14)$$

where t controls the magnitude of depth perturbation. We empirically set $t = 0.15$. Note that for focused objects exhibiting significant depth disparities, a larger t can be employed to better preserve geometric details. This formulation maintains the internal relative spatial hierarchy of the object, thus eliminating the unnatural visual effect arising from fully uniform depth assignment.

Directly using $\mathbf{M}_v$ in Eq. (4) might cause artifacts at the edge region. To solve this problem, we apply dual-band Gaussian smoothing to create soft alpha transitions $\mathbf{M}'_v = G_{r_{\text{in}}, s_{\text{in}}}(\mathbf{M}_v) \cdot G_{r_{\text{out}}, s_{\text{out}}}(\mathbf{M}_v)$, where $G_{r,s}$ denotes Gaussian

TABLE I
QUANTITATIVE SEGMENTATION RESULTS ON NVOS DATASET.

Metric	NVOS [24]	ISRF [28]	SA3D [29]	SAGD [30]	SA3D-GS [31]
mIoU (%)	70.12	83.85	90.31	90.41	90.72
mAcc (%)	92.01	96.42	98.21	98.21	98.31
Metric	SAGA [32]	OmniSeg3D [33]	COB-GS [23]	-	BRG-Bokeh
mIoU (%)	90.90	91.70	92.01	-	92.97
mAcc (%)	98.30	98.40	98.48	-	98.64

TABLE II
ABLATION STUDY OF OUR METHOD ON NVOS DATASET.

	$\mathcal{L}_{\text{bg}}$	$\mathcal{L}_{\text{edge}}$	$\mathcal{L}_{\text{depth}}$	SAM3	mIoU (%)	mAcc (%)
#1					92.01	98.48
#2	✓				92.58	98.57
#3	✓	✓			92.78	98.60
#4	✓	✓	✓		92.89	98.64
#5	✓	✓	✓	✓	92.97	98.64

filtering with radius r and strength s applied to inner and outer boundary bands. We set $r_{\text{in}} = 3$, $r_{\text{out}} = 3$, $s_{\text{in}} = 1.0$, $s_{\text{out}} = 1.0$. This eliminates edge artifacts while maintaining object silhouette integrity.

The final bokeh image $\mathbf{B}_v$ can be generated by $\mathbf{B}_v = \phi_{\text{Bokeh}}(\mathbf{I}_v, \mathbf{D}''_v, \mathbf{M}'_v, z_f^*, k)$.

IV. EXPERIMENT

A. Setup

Datasets. We follow COB-GS [23] and utilize the NVOS [24] benchmark, where multi-view consistent masks are generated by annotating prompt points on the reference view and propagating them using SAM3 [21]. For comprehensive qualitative evaluation, we also employ scenes from LLFF [25], MIP-360 [26], T&T [27], and DoF-NeRF [14].

Baselines. For 3D segmentation, we employ 8 state-of-the-art (SOTA) segmentation models, including NVOS [24], ISRF [28], SA3D [29], SAGD [30], SA3D-GS [31], SAGA [32], OmniSeg3D [33], and COB-GS [23]. For the evaluation of bokeh rendering performance, we utilize four representative methods, namely BRViT [34], DMSHN [35], BokehMe [36], and Dr.Bokeh [6]. To assess the consistency of bokeh effects, we adopt three controllable depth-of-field (DoF) methods based on 3D representations, specifically NeR-Focus [15], DOF-GS [17], and DoF-Gaussian [16].

Criteria. We adopt the mean Intersection over Union (mIoU) and mean accuracy (mAcc) to evaluate the performance of 3D segmentation tasks. For the quantitative assessment of bokeh rendering quality, we utilize four no-reference image quality evaluation metrics, namely the Natural Image Quality Evaluator (NIQE) [37], Perception-based Image Quality Evaluator (PIQE) [38], Hyper Image Quality Assessment (HyperIQA) [39], and Multi-scale Image Quality Transformer (MUSIQ) [40]. Additionally, we incorporate CLIP-IQA [41], a specialized no-reference metric designed to quantify the alignment between the generated image and predefined text prompts. Specifically, we use the prompts “natural bokeh”

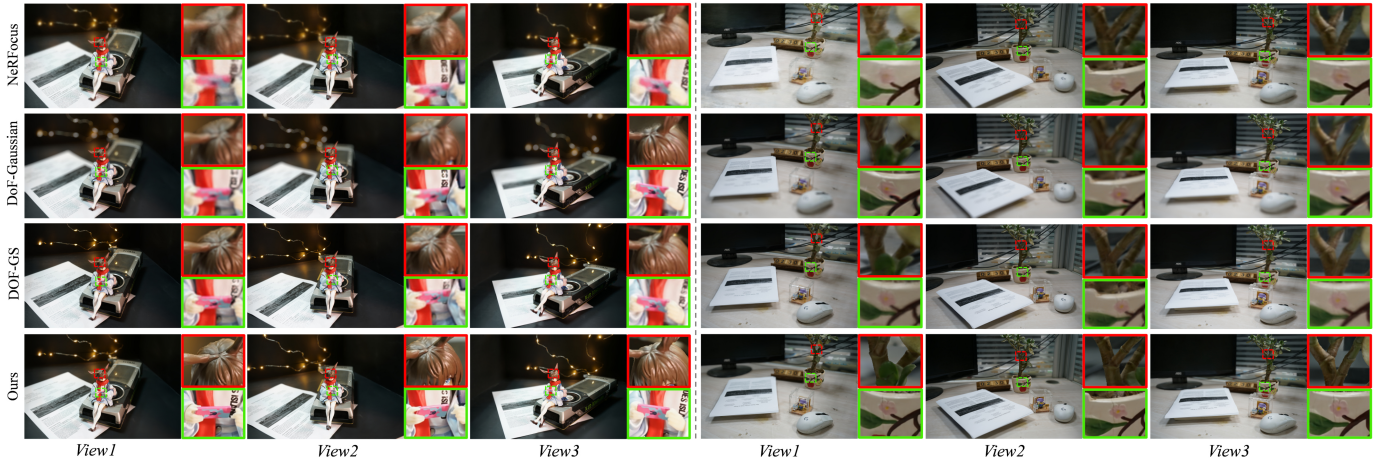


Fig. 2. Visual comparison of bokeh rendering results on “amiy” and “plant” scenes of DoF-NeRF dataset. The doll and the potted plant are in-focus. We manually adjust the focal distance of baselines to center on the target objects and maintain consistent aperture parameters.

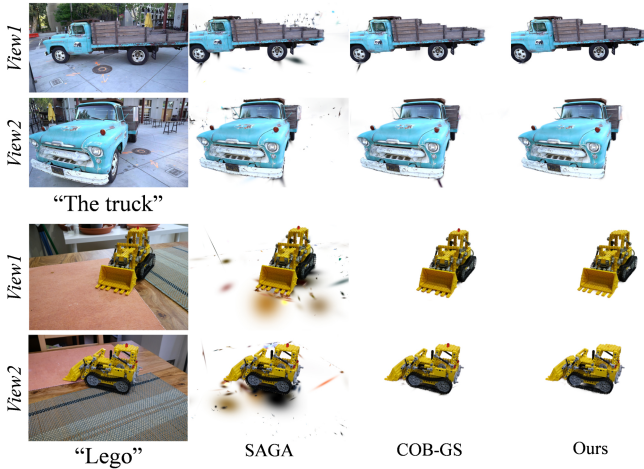


Fig. 3. Segmentation results on “truck” and “kitchen” scenes.

and “artificial bokeh” to differentiate between renderings with realistic visual effects and those with synthetic artifacts.

B. Segmentation Results

Table I shows the results of different 3D segmentation methods on NVOS dataset. We can find that 3D Gaussian-based methods (e.g., SAGD [30], SA3D-GS [31], SAGA [32], and COB-GS [23]) outperform NeRF-based methods (e.g., NVOS [24], ISRF [28], SA3D [29] and OmniSeg3D [33]). Among them, COB-GS [23] achieves the best performance with 92.01% and 98.48% mIoU and mAcc, respectively. Compared with COB-GS [23], our method achieves 0.96% and 0.16% relative mIoU and mAcc improvements. We also conduct ablation studies of our method in Table II. Our results demonstrate that every individual component plays a beneficial role in boosting overall performance. Tables I and II provide concrete evidence to validate the efficacy of our method in 3D segmentation tasks.

TABLE III
QUANTITATIVE BOKEH RESULTS ON THE NVOS DATASET.

Method	NIQE↓	PIQE↓	Hyper-IQA↑	MUSIQ↑	CLIP-IQA↑
BRViT [34]	6.54	77.31	0.3918	55.19	0.4510
DMSHN [35]	6.12	75.65	0.3983	50.53	0.4809
BokehMe [36]	6.37	56.59	0.3931	49.94	0.4585
Dr.Bokeh [6]	6.61	65.93	0.4087	54.48	0.4931
BRG-Bokeh	5.59	55.01	0.5807	69.62	0.5668

We further present qualitative comparisons on the “truck” (T&T) and “kitchen” (MIP-360) scenes in Fig. 3. The results reveal that feature-based methods (e.g., SAGA [32]) tend to exhibit inadequate semantic consistency, attributed to the ambiguity inherent to high-dimensional semantic features. Meanwhile, purely mask-based methods (e.g., COB-GS [23]) typically relying solely on global mask supervision, fail to fully leverage detailed mask information, rendering them susceptible to fine-grained noise along object edges and in background regions. In contrast, our method effectively alleviates these limitations: as illustrated in Fig. 3, it suppresses spurious tiny Gaussians along object boundaries and eliminates erroneous background artifacts, resulting in significantly sharper edges and cleaner segmentation masks.

C. Bokeh Result

Table III reports quantitative bokeh results on the NVOS dataset. Existing baselines (e.g., BokehMe [36], Dr.Bokeh [6]) exhibit metric trade-offs: some perform well in distortion (e.g., BokehMe’s PIQE) but lag in perceptual quality, while others (e.g., DMSHN [35]) show stronger perceptual scores but higher distortion. Our BRG-Bokeh outperforms all baselines across all metrics: it attains the lowest NIQE (5.59) and PIQE (55.01) (reducing bokeh distortion) and the highest Hyper-IQA (0.5807), MUSIQ (69.62), and CLIP-IQA (0.5668) (boosting perceptual quality/consistency). These results verify BRG-Bokeh’s superior bokeh performance on NVOS.

Fig. 2 shows some typical bokeh results on “amiya” and “plant”. NeRFocus [15] exhibits insufficient sharpness in focused regions, accompanied by significant degradation in rendering quality. DoF-Gaussian [16] generates prominent artifacts along focus boundaries, which undermines visual coherence. While DOF-GS [17] preserves overall scene clarity, it produces minimally expressive bokeh effects and thus fails to emphasize the target subject via a distinct shallow depth of field. In contrast, our method generates natural and aesthetically appealing bokeh effects without relying on defocus supervision. Furthermore, leveraging accurate 3D segmentation, our approach guarantees strictly consistent focus on the target object across diverse viewpoints, thereby validating its robustness in multi-view scenarios. Please refer to the Supplementary Material for additional experimental results and detailed ablation studies.

V. CONCLUSION

In this paper, we have proposed a novel framework to generate multi-view consistent bokeh effects for foreground objects in 3D scenes. Our framework leverages Depth Anything v3 for monocular depth maps and SAM3 for text-prompt-driven masks as ground truth, enabling the learned 3DGS to simultaneously generate RGB images, depth maps, and spatially consistent masks of the in-focus object across multiple views. We also develop a tailored multi-view bokeh rendering method that reassigns depth values to in-focus pixels and applies dual-band Gaussian smoothing to mitigate edge artifacts. Our approach surpasses existing 3D segmentation methods in mask quality, and achieves flexible, natural, and multi-view consistent bokeh rendering effects.

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